🧪 PHYSICAL PROPERTIES OF MATERIALS — FULL MIND MAP

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📄 Topics in this chapter:

1. States of Matter & Elasticity

- Liquid ka fixed volume hota hai lekin fixed shape nahi hota; kerosene iska example hai. Solid ka shape + volume fixed hota hai, jabki gas ka neither fixed shape nor fixed volume. *[Q1]*
- Young's modulus tensile stress/tensile strain ka ratio hai. Steel ka Young's modulus rubber se kaafi zyada hota hai. Coil/spring ki stretching mainly shear modulus (modulus of rigidity) se related hoti hai. *[Q2]*
- Same material ke wire ke liye elongation $\Delta L = FL/(YA) \propto L/d^2$. Isliye lamba aur patla wire same load par zyada stretch karega; source options me 3 m length, 1.5 mm diameter wala wire maximum elongate hota hai. *[Q13]*

2. Surface Tension — Drops, Oil & Soap Bubbles

- Surface tension liquid surface ko minimum possible area lene ki tendency deta hai; given volume ke liye sphere ka surface area minimum hota hai, isliye water/rain drops approximately spherical hote hain. *[Q3, Q4, Q5, Q6]*
- Oil ka surface tension water se kam hone par oil water surface par spread karta hai. Kerosene ka water par float karna surface tension se nahi, balki lower density se explain hota hai. *[Q7, Q8]*
- Soap bubble ke andar pressure atmosphere se greater hota hai; soap bubble ke liye excess pressure $\Delta P = 4T/r$. Smaller bubble ka internal pressure larger bubble se higher hota hai. *[Q9, Q10]*
- Hydrogen-filled flexible balloon altitude badhne par lower external atmospheric pressure ki wajah se expand karta hai. *[Q11]*
- Soap bubble par charge dene se like charges surface par repel karte hain, outward electrostatic pressure badhta hai aur equilibrium radius increase karta hai. *[Q12]*

3. Capillarity — Wicks, Blotting Paper & Plants

- Capillary action narrow pores/tubes me surface tension + adhesion/cohesion ki wajah se liquid rise/fall karne ki tendency hai. Kerosene wick aur blotting paper classic examples hain. *[Q14, Q16]*
- Plants me roots se leaves tak water rise sirf capillarity se nahi hota; transpiration pull, cohesion-tension aur root-related effects bhi important hain. *[Q14, Q15]*
- Jurin's law: $h = 2T\cos\theta/(r\rho g)$. Capillary rise T se directly aur tube radius तथा density se inversely related hota hai (other factors same). *[Q17]*

> ### ⚠️ EXAM TRAP — Q14 & Q17 — Capillarity Questions Need Conditions
> ▸ Straw se soft drink peena capillary action ka standard example nahi hai; mouth se pressure difference create karke liquid upar aata hai. Isliye Q14 me statement 2 ko exclude karna correct hai.
>
> ▸ Capillary height sirf surface tension par depend nahi karti: $h = 2T\cos\theta/(r\rho g)$. Q17 ka "higher T ⇒ higher rise" conclusion tabhi safe hai jab radius, contact angle aur density comparable/assumed same hon.

4. Fluid & Atmospheric Pressure

- Lake ke bottom se top ki taraf aate air bubble par hydrostatic pressure kam hota hai, isliye Boyle's law ke according bubble volume/size increase karta hai. *[Q18]*
- High altitude par external atmospheric pressure kam hota hai; fountain pen ke andar trapped air comparatively higher pressure se ink ko bahar push kar sakti hai. *[Q19]*
- Hydraulic brakes Pascal's law par work karte hain: confined incompressible fluid me applied pressure all directions me equally transmit hota hai. *[Q20]*
- Barometer atmospheric pressure measure karta hai. Sudden sharp fall generally approaching storm/low-pressure disturbance ka indicator mana jata hai. *[Q21]*
- Torricelli relation $v = \sqrt{2gh}$: free surface se deeper hole par efflux speed greater hoti hai. *[Q22]*

⚠ EXAM TRAP — Q22 — "Which Hole Throws Water Farthest?"

▸ Torricelli law deeper hole ko higher exit speed deta hai, but horizontal range ground level/exit height par bhi depend karti hai.

▸ Source key bottom hole (1) deta hai. Agar exam figure me holes ki heights/landing geometry materially differ ho, range ke liye projectile motion bhi consider karo; source stem ko exactly read karo.

5. Viscosity & Streamline Flow

- Laminar flow in a cylindrical pipe me no-slip condition ki wajah se wall ke paas fluid speed minimum (≈0) aur central axis par maximum hoti hai; velocity profile parabolic hota hai. *[Q23]*
- Liquids ki viscosity temperature badhne par generally decrease karti hai because molecular mobility increases and cohesive effects weaken. *[Q24, Q27]*
- Honey given common options me water, air, blood, alcohol/gasoline se much more viscous hai; viscosity flow ke resistance/internal friction ka measure hai. *[Q25, Q26]*

6. Density as an Intensive Property

- Substance ki amount badhane se mass, weight aur volume badh sakte hain, lekin same conditions par density $\rho = m/V$ unchanged rehti hai; density intensive property hai. *[Q28]*

7. Anomalous Expansion of Water

- Water 0°C–4°C range me anomalous behaviour dikhata hai: 0→4°C heat karne par volume decreases aur density increases; 4°C par volume minimum aur density maximum hoti hai. *[Q29, Q32, Q34, Q35, Q36]*
- 4°C se upar water normal liquids ki tarah heating par expand karta hai. Isliye 0→10°C me volume pehle decrease, phir increase hota hai. *[Q32]*
- 9°C→3°C cooling me volume pehle 4°C tak decrease karta hai, phir 4→3°C me increase karta hai. *[Q33]*
- Winter me surface water 0°C par freeze ho sakta hai while denser 4°C water bottom par liquid rehta hai. Ice/top cold layer insulation provide karti hai, isliye aquatic life survive kar sakti hai. *[Q29, Q30, Q31]*
- Maximum density of water 4°C ≈ 277 K par hoti hai. *[Q34, Q35, Q36]*

8. Gas Laws & Liquefaction

- Avogadro's law: same temperature and pressure par equal volumes of gases equal number of molecules contain karte hain; $V \propto n$. *[Q37]*
- Gas liquefaction ke liye low temperature + high pressure favorable hota hai: kinetic energy reduce hoti hai aur molecules closer aate hain. *[Q38]*
- Gas molecules ki average separation container volume increase karne ya some gas leak hone par increase ho sakti hai; fixed volume me more gas/additional compression separation reduce karta hai. *[Q39]*
- Charles's law (constant pressure): V/T = constant. Half-volume gas at 27°C = 300 K ko original volume tak expand karne ke liye temperature 600 K = 327°C chahiye. *[Q40]*

9. Relative Density & Floating — Iron, Mercury, Ships

- Iron water se denser hai, isliye solid iron ball/nail water me sink karta hai; mercury iron se much denser hai, isliye iron mercury par float kar sakta hai. *[Q41, Q42]*
- Iron needle normally sink karti hai because iron density > water. Hollow iron ship ka average density (metal + enclosed air) water se sufficiently low hota hai aur displaced water ka buoyant force ship ko float karata hai. *[Q43]*
- Carefully placed small iron needle water surface tension ki supporting force se surface par remain kar sakti hai, despite iron being denser than water. *[Q44]*
- Archimedes principle: immersed body par upthrust = displaced fluid ka weight. Isliye submerged bucket lighter feel hoti hai; water se bahar aate hi buoyant support kam ho jata hai aur bucket heavier feel hoti hai. *[Q45]*
- Weightless balloon me 200 g water ho aur same-density water me fully immersed ho, to displaced water ka weight contained water ke weight ke equal hoga; net apparent weight zero. *[Q46]*
- Cloud droplets ka effective bulk density surrounding lower atmosphere ke comparison me low hota hai aur air currents/buoyancy unhe suspended rakhne me help karte hain. *[Q47]*

⚠ EXAM TRAP — Q47 — Why Clouds Stay Aloft

- ▸ "Low density" exam-level shorthand hai. Individual cloud droplets liquid water se dense hote hain, but bahut tiny size ki wajah se terminal settling speed low hoti hai and atmospheric updrafts/turbulence them suspended rakh sakte hain.
- ▸ Cloud ko ek solid low-density object ki tarah simple buoyancy se explain karna incomplete model hai.

10. Icebergs & Floating Ice

- Floating body ka submerged fraction ≈ ρ_{body}/ρ_{fluid}. Ice ki density water/seawater se lower hone ki wajah se major part submerged aur small fraction surface ke above hota hai. *[Q48]*
- Floating ice jitna water displace karti hai, melt hone par utna hi water volume contribute karti hai (weight-equivalent basis); pure floating ice melt hone par water level essentially unchanged rehta hai. *[Q49, Q50]*

⚠ EXAM TRAP — Q48 — Iceberg Fraction Above Sea Water

- ▸ Source answer 1/9 above surface deta hai. Exact fraction densities par depend karta hai: above fraction = 1 − $\rho_{ice}/\rho_{seawater}$.
- ▸ Typical density values use karne par visible fraction roughly one-tenth ke order ka aata hai, so PYQ me source-specific option/key follow karte waqt formula yaad rakho.

11. Temperature Effect on Buoyancy

- Water 4°C par maximum density rakhta hai. Temperature 100°C ki taraf badhne par density decrease hoti hai, isliye same immersed volume ke liye buoyant force reduce hota hai aur floating body ka larger fraction submerged hoga. *[Q51]*

⚠ EXAM TRAP — Q51 — "At 100°C the Body Will Sink" is Not Universally Guaranteed

- ▸ Water density heating par decrease hoti hai, so a floating body ko support karne ke liye greater submerged fraction chahiye.
- ▸ Body tabhi completely sink karegi jab uski density heated water ki density se greater ho. Sirf "it floated at 4°C" se sinking at 100°C logically guaranteed nahi hoti; source key is a simplified exam assumption.

12. Sea Water, Swimming & Ships

- Sea water dissolved salts ki wajah se river/fresh water se denser hota hai; greater density greater buoyant force provide karti hai, isliye sea me swimming easier feel hoti hai. *[Q52]*
- Ship river se sea me jaata hai to denser sea water same weight ko support karne ke liye less volume displace karta hai, so ship thoda rise karta hai. *[Q54, Q55]*

13. Apparent Weight in Different Media

- Apparent weight = true weight − buoyant force. Given media me hydrogen sabse low-density hai, so uska upthrust least aur body ka apparent weight highest hota hai. *[Q53]*